Research Statement

Towards a p-adic spectral geometry

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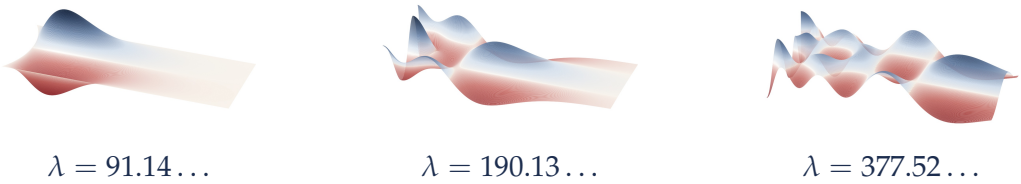
Overview

A recurring theme in modern number theory is that *arithmetic phenomena* first encountered analytically are explained by their *geometric nature*. Values of automorphic functions, meromorphic modular forms and Laplacian spectral patterns often reflect the underlying structures of moduli spaces and their associated motives.

A fundamental source of examples comes from Riemann surfaces $\Gamma\backslash\mathbb{H}$ obtained as a quotient of the upper-half plane $\mathbb{H} = \{z = x + iy : y > 0\}$ by Möbius transformations under a Fuchsian group $\Gamma \subseteq \mathrm{PSL}_2(\mathbb{R})$. The *Laplace-Beltrami* operator Δ is explicitly given by

$$\Delta = -y^2(\partial_x^2 + \partial_y^2). \tag{1}$$

This operator governs diffusion and wave propagation on a negatively curved space. On a hyperbolic surface, this captures the dynamics, the geometry and, when the surface is arithmetic, the number theory of the space. Its eigenfunctions $\Delta\varphi = \lambda\varphi$, called *Maaß forms*, are hyperbolic waves (Figure 1) and their nodal domains (see Figure 3) represent the geometry of sign changes of these waves.



$\lambda = 91.14\ldots$ $\lambda = 190.13\ldots$ $\lambda = 377.52\ldots$

Figure 1: Three Maaß cusp forms for $\mathrm{SL}_2(\mathbb{Z})$, with Laplace eigenvalues λ over the fundamental domain $\{|\mathrm{Re}\, z| \leq \frac{1}{2},\ |z| \geq 1\}$, with the cusp at the right.

In the arithmetic case, for example the *modular curve* $Y(1) = \mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$, these analytic objects unexpectedly acquire deep arithmetic meaning central in modern number theory:

1. special Maaß eigenforms conjecturally encode Galois representations and L-functions,
2. special values of automorphic Green kernels connect L-functions with heights and intersections of special algebraic cycles.

The first is an incarnation of the *Langlands philosophy* while the second lies behind the Gross–Zagier–Kolyvagin approach to the rank-one *Birch and Swinnerton-Dyer conjecture*. My research asks what is the analogue of this picture in p-adic geometry. Together, these projects form a programme toward a p-adic spectral geometry of modular and Shimura curves.

- **Project 1.1** studies *p-adic Green functions* on the Drinfeld upper half-plane, develops explicit Nekovář *p-adic heights* of CM cycles on Shimura curves and proves a p-adic analogue of the averaged *Gross–Zagier algebraicity formula*.
- **Project 1.2** proves previously conjectured congruences between coefficients of meromorphic modular forms and the number of points of elliptic curves over finite fields. It studies *extension of filtered φ-modules* on modular curves comparing explicitly Frobenius on cohomology with U_p-operators.
- **Project 1.3** studies the *spectral analogies* between the Laplacian operator on Maaß forms and the U_p operator on *overconvergent modular forms*. It focuses on the zero distribution of eigenforms.

1 PhD work:
three p-adic approaches to analytic phenomena

1.1 p-adic heights on Shimura curves and higher p-adic Green functions [BH26]

[Joint preprint with Hazem Hassan, available here (49 pages)]

In joint work with Hazem Hassan, I developed a p-adic analogue of the higher Green functions $G_k(z,w)$, we realized them as explicit local p-adic heights of CM cycles and we proved that the average sum over all these cycles is the p-adic log of a rational number.

The value of higher Green functions at CM points has been conjectured in [GKZ87] to be equal to the *logarithm of an algebraic number*. This has sparked research in several directions and has recently been resolved in many cases by fairly different methods, attracting researchers both from a more analytic side [BEY21], [Via11] and [Li23], via the theory of Borcherds lifts and analytic Maaß forms, and from a geometric one [Zha97], [Mel08] and [BF25] using a more motivic approach. This latter approach gave us a natural template for a p-adic construction using p-adic heights in the sense of [Nek93]. If furthermore we work on Drinfeld p-adic upper half-plane (Figure 2 left), we can explicitly describe these functions in terms of p-adic analytic analogues of *Legendre functions*.

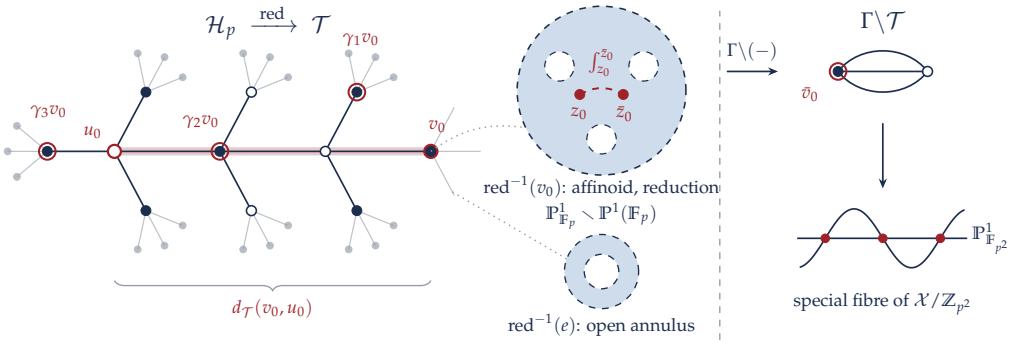


Figure 2: The Čerednik–Drinfeld uniformisation at a prime p. *Left.* Drinfeld's upper half plane $\mathcal{H}_p = \mathbb{P}^1_{\mathbb{C}_p} \smallsetminus \mathbb{P}^1(\mathbb{Q}_p)$. *Right.* $X_{\mathbb{Q}_{p^2}} \cong \Gamma\backslash\mathcal{H}_p$. The finite graph $\Gamma\backslash\mathcal{T}$ is the dual graph of the special fibre.

Local height formula

Fix a Shimura curve X admitting p-adic uniformization via Čerednik–Drinfeld (Figure 2) and fix a positive even integer k. We constructed extensions supported at CM points, where p is inert, producing *biextensions of the semistable p-adic Galois representation* and we studied the corresponding *filtered (φ, N)-modules*

$$V = H^1_{\text{ét}}(X_{\overline{\mathbb{Q}}}, \mathbb{L}_{k+2}) \qquad \leftrightarrow \qquad D_{\text{st}}(V_p) = H^1_{\text{dR}}(X_{\mathbb{Q}_p}, \mathcal{V}_{k+2}) \tag{2}$$

to which Nekovář's general local height pairing formalism applies. In the paper, we have extended Besser's [Bes02] [Bes22] and Besser–Zerbes's [BZ23] p-adic height approach from trivial to arbitrary symmetric-power coefficients, and we adapt Brown–Fonseca's generating-series computation of the archimedean primitive to the p-adic analytic setting.

Theorem 1 ([BH26]). *Let z_0, w_0 be two Heegner points on $X = \Gamma\backslash\mathcal{H}_p$ associated to two imaginary quadratic fields K_1, K_2. Associate to them two simple extensions $e_z, e_w \in H^1_{\text{st}}(\mathbb{Q}_p, V)$. The p-adic local height is given by*

$$h(e_z, e_w)_p = \text{Tr}_{\mathbb{Q}_{p^2}/\mathbb{Q}_p}\left(\int_{z_0}^{\bar{z}_0} \Psi_k(z, w_0)dz\right) = \text{Tr}_{\mathbb{Q}_{p^2}/\mathbb{Q}_p}\left(\sum_{\gamma\in\Gamma} Q_{(k-2)/2}(1 - C(\gamma z_0, w_0))\right) \tag{3}$$

where Ψ_k is an explicitly expressed meromorphic modular form analogue to the Poincaré series, Q_{k+2} are p-adic analogues of Legendre functions and C is a cross-ratio. The formula holds up to an explicit renormalization.

Analytic realization and algebraicity

The right hand side of formula (3) is denoted by $\mathrm{Tr}_{\mathbb{Q}_{p^2}/\mathbb{Q}_p}\left(G_k^{(p)}(z_0,w_0)\right)$ and represents the p-adic *higher Green function*. We can now act by the Hecke algebra on the global cohomology to make the biextension a simple extension of Tate motives, i.e. a *Kummer motive*. This yields the following theorem.

Theorem 2 ([BH26] A p-adic analogue of the Gross–Zagier algebraicity conjecture)**.** *Consider a non-zero finite linear combination of Hecke operators T which annihilates $H^1_{\text{ét}}(X,\mathbb{L}_{k+2})$. Let K_1,K_2 be imaginary quadratic fields with coprime discriminants. Then the sum over all Heegner points z_0,w_0 associated to K_1,K_2 of the evaluation of the higher Green's function is equal to*

$$\mathrm{Tr}_{\mathbb{Q}_{p^2}/\mathbb{Q}_p}\left(\sum_{z_0,w_0} G_k^{(p)}(z_0,Tw_0)\right) = C_{k,K_1,K_2}\log_p(\alpha_T) \tag{4}$$

for $\alpha_T\in\mathbb{Q}$ and an explicit constant C_{k,K_1,K_2}.

1.2 A p-adic cohomological approach to congruences of meromorphic modular forms

[Single-author preprint available on arXiv (12 pages)]

Arithmetic phenomenon

In order to explain the content of the paper and the main result we can start with the following phenomenon first observed and conjectured by Pengcheng Zhang in [Zha25]. Consider the following elliptic curve and the meromorphic modular form

$$C:\ y^2+xy=x^3-x^2-2x-1,\qquad F=\frac{E_4}{j+3375}=\sum_{n\geq 1}a_n(F)q^n.$$

Here E_4 is the weight four Eisenstein series and $q=e^{2\pi i\tau}$. For an ordinary prime $p>3$, let $u_p(C)$ be the unit root of $T^2-a_p(C)T+p$, where $a_p(C)=p+1-\#C(\mathbb{F}_p)$. Then Zhang conjectured

$$\boxed{a_{np^\ell}(F)\equiv u_p(C)^2 a_{np^{\ell-1}}(F)\pmod{p^{3\ell}},\qquad n,\ell\geq 1.} \tag{5}$$

This surprising phenomenon links the U_p-action on meromorphic modular forms and Frobenius eigenvalues on cohomology of the elliptic curve. The paper proves and explains this type of congruence.

Theorem 3 ([Bor26])**.** *Fix $E/\mathbb{Q}$ an elliptic curve with good reduction at p and k a positive even integer. There exist meromorphic modular forms $f_0,\dots,f_k$ of weight $k+2$ with poles at $j(E)$ such that for all integers $m,l\geq 1$*

$$a_{mp^\ell}(f_i)\ \equiv\ \alpha^i\beta^{k-i}\cdot a_{mp^{\ell-1}}(f_i)\pmod{p^{(k+1)\ell}}\quad i=0,\dots,k$$

where α,β are the roots of the crystalline Frobenius polynomial on $H^1_{\mathrm{dR}}(E,\mathbb{Q}_p)$.

p-adic geometric toolkit

It is key to consider the *rigid cohomology* on modular curves, the theory of *overconvergent modular forms* and the *p-adic geometry* of the modular curve. Using an idea of Coleman [Col96] involving Katz–Lubin *canonical subgroup*, we can give an explicit description of the action of Frobenius on the de Rham cohomology group of a modular curve X with a system of coefficients $\mathcal{H}$ in terms of the Hecke endomorphism U_p on space of meromorphic modular forms

$$\boxed{\text{Frob}\ \circlearrowleft\ H^1_{\mathrm{dR}}(X-D,\mathrm{Sym}^k\mathcal{H})}\quad\longleftrightarrow\quad\boxed{M^D_{k+2}/\theta^{k+1}M^D_{-k}\ \circlearrowleft\ U_p,V_p.} \tag{6}$$

The core of the project is translating the initial congruence into a modular machinery where the study of extensions of filtered φ-modules by simple pieces ties the analytic and geometric data. As a byproduct of this proof of the conjecture of Zhang, this work sheds some light on a modular and explicit action of Frobenius on cohomology that can be relevant for recent work in *explicit quadratic Chabauty area* for developing algorithms that do not depend on models of the curves [Ren25].

1.3 Spectral aspects of p-adic Maaß forms

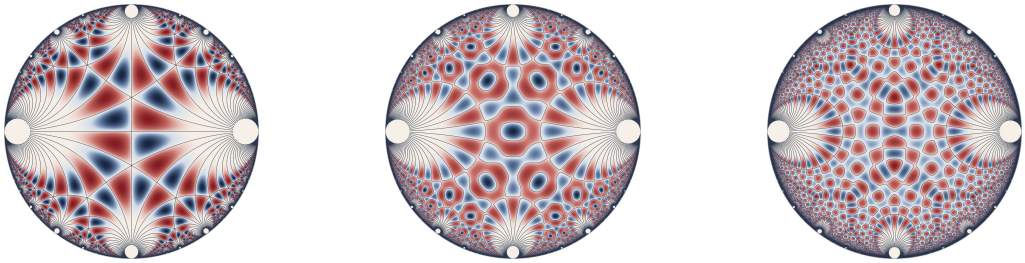
[Ongoing joint work with Jan Vonk]

A spectral dictionary

A theme running through my thesis is that the analytic Laplacian Δ seems to have a counterpart in p-adic geometry. A striking instance of this, going back to suggestions of Blasius and Calegari, is the analogy

$$\left\{ \begin{array}{c} \text{Maaß forms } \mathcal{M}_\infty^{\mathrm{an}} \\ \Delta = -y^2(\partial_x^2 + \partial_y^2) \end{array} \right\} \quad \longleftrightarrow \quad \left\{ \begin{array}{c} \text{Overconvergent forms } \mathcal{M}_p^{\mathrm{an},r} \\ \text{“}\Delta\text{”} = \log|U_p| \end{array} \right\} \tag{7}$$

between the *hyperbolic Laplacian* acting on *Maaß forms* and the *Atkin operator* U_p acting on the space $M_0^\dagger(r)$ of *r-overconvergent modular functions*, under which the Selberg spectral gap corresponds to the smallest positive slope of U_p, Weyl's law to the asymptotics of the slope sequence [Wan98], and the nodal set (Figure 3) of a Maaß eigenfunction [HR92] to the zero locus of an overconvergent one.



$\lambda = 91.14\ldots$ $\lambda = 190.13\ldots$ $\lambda = 377.52\ldots$

Figure 3: Three Maaß cusp forms for $\mathrm{SL}_2(\mathbb{Z})$, with Laplace eigenvalues λ. The dark lines represent the nodal sets.

Archimedean equidistribution of such zero loci is a theorem for holomorphic forms [Rud05]; on the p-adic side the patterns that have been observed come with no geometric mechanism, and supplying one is the aim of this project.

Exact zero radii

Let p be a prime for which $X_0(p)$ has genus zero, the explicit decompositions of U_p obtained by Buzzard–Calegari [BC05] and Loeffler [Loe07] imply that the eigenfunctions f_n on $M_0^\dagger(r)$, ordered by increasing slope have exactly n zeroes on X_r. We have computed these zero divisors for $p = 2, 3, 5, 7, 13$, and find that their radii not only accumulate at $v(A) = 1/(p+1)$, with A a lift of the Hasse invariant, but appear to obey an exact law: on the component $n \equiv 1 \pmod{p-1}$ we conjecture

$$\boxed{\mathrm{Div}\left(f_{p^n}\right) \;=\; [\infty] \;+\; \sum_{k=0}^{n-1} p^k(p-1) \text{ roots of radius } r_k\,,} \qquad r_k \;=\; \frac{1}{p+1} \;-\; \frac{1}{\mathrm{lcm}(12,\, p+1)\; p^k}. \tag{8}$$

Calegari observed [Cal13] that f_{2^i} converges to the 2-*depletion* $E_0^{[2]}$ of the weight zero Eisenstein series;

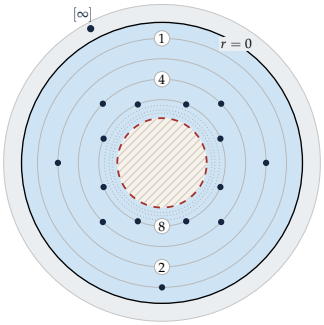


Figure 4: The zero divisor of the p-depleted Eisenstein series $E_0^{[p]} = \log_p\big(V(h)/h^p\big)$. For $p = 2$ the radii are $\frac{1}{4}, \frac{7}{24}, \frac{5}{16}, \frac{31}{96}, \cdots \to \frac{1}{3}$; the first n circles contain $\mathrm{Div}(f_{2^n})$.

we conjecture in general that for p odd

$$\lim_{n\to\infty} f_{p^n} \;=\; \sum_{(p,m)=1} \chi^{(p-1)/2}(m)\,\sigma_{-1}(m)\,q^m, \tag{9}$$

where χ is the Teichmüller character, so that the limit is the *Eichler integral* of a weight two Eisenstein series with quadratic nebentypus, an infinite slope form of the kind studied in [BGK12]. These limits appear arithmetically rich, and we are currently studying them.

Geometric mechanism

Let $h(z) = (\Delta(pz)/\Delta(z))^{\frac{1}{p-1}}$ be the Hauptmodul, we can then express $E_0^{[p]} = \log_p\big(V(h)/h^p\big)$. The zeroes of the limit function can be explicitly characterized by studying $V(h)/h^p$. We evaluate $V(h)/h^p$ on the *formal group of the universal elliptic curve* using Kato's theta functions [Kat04] together with the canonical subgroup theory of Katz [Kat73] and Lubin [Lub79]. This identifies the accumulation radius as the threshold at which the canonical subgroup of order p^2 ceases to exist. Determining the exact valuation of $V(h)/h^p - 1$ as a function of $v(A)$, which is work in progress, would locate every zero of the limit function.

2 Future programme: toward a p-adic spectral geometry

The three projects above are connected by more than a shared toolbox. The dichotomy that organises the spectral picture of §1.3 is remarkably similar to the one that separates the archimedean from the non-archimedean local height contributions in [GZ86] and [GKZ87], which are precisely the objects that §1.1 computes p-adically. That one splitting should govern both the analytic behaviour of U_p and the decomposition of a p-adic height pairing is unlikely to be an accident, and making it structural is the direction in which I would most like to push this circle of ideas.

2.1 A higher-weight p-adic Gross–Zagier formula

The formula of §1.1 evaluates local heights of CM cycles; the natural next step is to relate their global counterpart to derivatives of p-adic L-functions, in the higher weight setting and for the quaternionic curves in the spirit of the archimedean results of [Zha97]. Because the expression we obtained is a Coleman integral of an explicit Poincaré series, a formula of this shape would come with a means of computing the relevant derivatives, and not only with an existence statement. Several p-adic approaches have appeared in the literature, Perrin-Riou [PR87] and Nekovář [Nek95] developed the p-adic counterpart of classical and higher weight Gross–Zagier formula in the modular curve case. More recently, Disegni [Dis17, Dis23] developed the Shimura curve case for weight two. Our work will then sit in the higher weight Shimura case with a p-adic analytic perspective. We also

believe that the explicit analytic expression, not available in the modular curve case, should shed new insights on the p-adic formula.

The dichotomy provided by [GZ84] between heights and analytic families of Eisenstein series has fruitfully sparked p-adic analogies on Shimura curves. The work of Daas [Daa25] proves the log rationality of Theta functions on Shimura curves using Galois deformation arguments. Our work can then recognize the Theta functions appearing as the p-adic local height contribution. This naturally leads to possible higher weight generalization using the method of Rankin–Cohen brackets and our analytic formula would supply the higher Green function counterpart.

2.2 Zero locus equidistribution for the U_p-operator

Beyond the zero divisor conjecture of §1.3, the dictionary with Maaß forms predicts asymptotic statements: a Weyl law for the slope sequence, and equidistribution of the zero loci as the slope grows, the p-adic counterpart of [Rud05]. My conjecture describes exact radii for one family at small genus; the goal is the corresponding asymptotics, uniformly in the prime and the level, where the geometry of the modular curve at infinite level provides tools with no archimedean analogue.

We have developed code to compute with high precision the Newton polygons attached to each U_p eigenform. This has provided tables of data that seems to indicate clustering of eigenforms indexed by Galois representations where the zeroes distribute according to a *ghost power series* combinatorially easier to describe. This translates the theme of the Ghost conjectures [BP19, LTXZ26] to the computation of zeroes on the supersingular annulus.

2.3 Explicit Frobenius and algorithms

The description of the Frobenius action obtained in §1.2 is explicit enough to compute with, and I intend to develop it into algorithms that require no choice of model for the curve, continuing the line of [Ren25]. This is the most computational strand of my work, and the one in which the code I have written for Frobenius on de Rham cohomology and for Gauss–Manin connections in families should be directly useful.

2.4 Meromorphic modular forms and CM elliptic curves

The case of CM elliptic curves can provide stronger and explicit statements for the congruences explored in Project 1.2. For example, the motivic decomposition of this curves propagates to the various p-adic realizations providing more structure compatible with Frobenius and filtration. A further explicit analysis of the description of the filtration of the meromorphic modular forms following [BF25] gives a direct computable space to work with.

2.5 Long-term synthesis

Taken together, these directions aim to identify the geometric structures that replace the *Laplace operator in p-adic geometry*. The long-term objective is a framework in which U_p-spectra, Frobenius actions and p-adic height pairings appear as different realizations of a common arithmetic spectral theory. Such a framework should connect explicit computations on modular curves with global questions concerning cycles and p-adic L-functions.

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